

1 Multipath Propagation, Fractional Delay, and Intersymbol Interference in Over-the-Air Computation

While [6] established the foundational premise of Over-the-Air (OTA) computing under an idealized, flat-fading Additive White Gaussian Noise (AWGN) channel [6], [8], practical wireless propagation environments diverge substantially from this benchmark [7], [10]. Electromagnetic waves emitted simultaneously by distributed End Device (ED)s propagate across diverse physical paths, undergoing reflection, diffraction, and scattering off terrain, obstacles, and structural boundaries. This phenomenon introduces multipath dispersion and unresolvable fractional time delays across arriving wavefronts. In conventional point-to-point digital communications, such channel-induced dispersion causes adjacent symbols to overlap in time, creating Inter-Symbol Interference (ISI). In OTA networks, where computation is executed natively through analog waveform superposition in the air, the implications of ISI are profoundly more severe: dispersion destroys the precise temporal alignment required for nomographic multi-access superposition, directly corrupting the spatial aggregation arithmetic [3], [5].

This chapter presents a comprehensive theoretical formulation, algorithmic discrete-time modeling, block-level Simulink architecture, and experimental diagnostic framework for analyzing OTA computation under multipath propagation and fractional delay. We derive the closed-form mathematical expression for the resulting irreducible Mean Squared Error (MSE) error floor, demonstrate how it scales quadratically with the multipath echo gain (g^2), evaluate the sub-sample fidelity of polynomial Farrow fractional delay interpolation, and validate the system dynamics across multi-rate sample-based Simulink architectures, animated eye patterns, and constellation diagrams.

1.a Continuous-Time Formulation and Multipath Channel Model

Consider an Over-the-Air computation network comprising K distributed sensor nodes deployed across an environment (such as an agricultural sensory grid), concurrently transmitting localized scalar measurements to a central Fusion Center (FC). Each sensor node $k \in \{1, \dots, K\}$ generates a discrete sequence of zero-mean, uncorrelated information symbols $x_k[m]$ with symbol duration T_s and statistical variance:

$$\mathbb{E}[x_k[m]] = 0, \quad \mathbb{E}[x_k[m]x_{k'}[m']] = \sigma_x^2 \delta_{k,k'} \delta_{m,m'} \quad (1)$$

where $\delta_{i,j}$ denotes the Kronecker delta function. In uncoded analog Pulse Amplitude Modulation (PAM), each device directly maps its local sensor reading to the amplitude of a transmitted continuous-time pulse. Let $p(t)$ denote an identical Root Raised Cosine (RRC) pulse-shaping filter implemented at each transmitter, possessing roll-off factor $\alpha \in [0, 1]$ and unit energy normalization:

$$\int_{-\infty}^{\infty} p^2(t) dt = 1 \quad (2)$$

The continuous-time baseband waveform radiated by the k -th device is expressed as:

$$s_k(t) = \sum_{m=-\infty}^{\infty} x_k[m] p(t - mT_s) \quad (3)$$

By linear electromagnetic superposition over the shared wireless medium, the aggregate transmitted waveform arriving over the line-of-sight path is:

$$s(t) = \sum_{k=1}^K s_k(t) = \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] p(t - mT_s) \quad (4)$$

1.a.i Two-Ray Multipath Channel Profile

To rigorously isolate the fundamental degradation induced by multipath dispersion without obscuring the underlying mechanics behind complex fading distributions, the propagation medium is modeled as a two-ray multipath channel. The channel consists of a dominant direct path and an unresolved secondary specular reflection (echo) characterized by an attenuation gain $g \in [0, 1]$ and a fractional time delay $\tau \in \mathbb{R}^+$:

$$h(t) = \delta(t) + g \delta(t - \tau) \quad (5)$$

where $\delta(\cdot)$ represents the Dirac delta distribution. Expressing the delay relative to the symbol interval yields the normalized fractional delay $\tau_{\text{norm}} = \tau/T_s$. The continuous-time signal arriving at the receive antenna of the FC prior to noise corruption is governed by the linear continuous convolution:

$$r(t) = (s * h)(t) + w(t) = s(t) + g s(t - \tau) + w(t) \quad (6)$$

where $w(t)$ denotes additive white Gaussian noise with two-sided power spectral density $N_0/2$.

1.a.ii Receiver Matched Filtering and Symbol Strobing

At the FC, the received waveform $r(t)$ passes through a matched filter identical to the transmitter pulse shape, $p^*(-t) = p(t)$. By design, the continuous-time convolution of two matched RRC filters yields the standard Nyquist Raised Cosine (RC) pulse $r_c(t) = (p * p)(t)$ [2], [7]:

$$r_c(t) = \text{sinc}\left(\frac{t}{T_s}\right) \frac{\cos\left(\frac{\pi \alpha t}{T_s}\right)}{1 - \left(\frac{2\alpha t}{T_s}\right)^2} \quad (7)$$

The output of the matched filter before sampling is given by:

$$y(t) = (r * p)(t) = \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] r_c(t - mT_s) + g \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] r_c(t - mT_s - \tau) + \tilde{w}(t) \quad (8)$$

where $\tilde{w}(t) = (w * p)(t)$ represents the filtered Gaussian noise process.

Sampling $y(t)$ at the nominal symbol strobe instants $t = nT_s$ yields the discrete-time observation sequence:

$$y[n] = \sum_{k=1}^K x_k[n] r_c(0) + \sum_{k=1}^K \sum_{m \neq n} x_k[m] r_c((n - m)T_s) + g \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] r_c((n - m)T_s - \tau) + \tilde{w}[n] \quad (9)$$

By the first Nyquist criterion for zero ISI, the direct-path RC pulse satisfies $r_c(0) = 1$ and $r_c(lT_s) = 0$ for all non-zero integers $l \neq 0$. Consequently, the direct-path summation collapses cleanly to the desired instantaneous sum, whereas the multipath echo introduces non-vanishing cross-symbol leakage:

$$y[n] = \underbrace{\sum_{k=1}^K x_k[n]}_{\text{Desired Superposition}} + \underbrace{g \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] r_c((n - m)T_s - \tau)}_{\text{Multipath-Induced Intersymbol Interference}} + \tilde{w}[n] \quad (10)$$

To estimate the true spatial arithmetic mean of the distributed sensors, $\mu[n] = \frac{1}{K} \sum_{k=1}^K x_k[n]$, the FC applies a standard post-scaling factor of $1/K$:

$$\hat{\mu}[n] = \frac{y[n]}{K} = \mu[n] + g \sum_{m=-\infty}^{\infty} \mu[m] r_c((n - m)T_s - \tau) + \frac{\tilde{w}[n]}{K} \quad (11)$$

Eq. (11) explicitly demonstrates the fundamental failure mode of analog Over-the-Air computing in dispersive channels: even when the line-of-sight path is perfectly synchronized, the fractional multipath echo smears previous and subsequent spatial averages $\mu[m]$ into the current estimation strobe n .

1.b Analytical Derivation of the Multipath Error Floor

To quantify the performance boundary of OTA computation under multipath interference, we examine the instantaneous computation error at the FC, defined as $e[n] = \hat{\mu}[n] - \mu[n]$:

$$e[n] = g \sum_{m=-\infty}^{\infty} \mu[m] r_c((n-m)T_s - \tau) + \frac{\tilde{w}[n]}{K} \quad (12)$$

The overall computation MSE is given by the statistical expectation of the squared error:

$$\text{MSE}(g, \tau, \text{SNR}) = \mathbb{E}[e^2[n]] \quad (13)$$

Because the information symbols $x_k[m]$ and the additive channel noise $\tilde{w}[n]$ are statistically independent and mutually uncorrelated, the cross-terms vanish under expectation, yielding a complete decoupling between the thermal noise variance and the multipath interference term:

$$\text{MSE}(g, \tau, \text{SNR}) = \text{MSE}_{\text{noise}}(\text{SNR}) + \text{MSE}_{\text{floor}}(g, \tau) \quad (14)$$

1.b.i Closed-Form Asymptotic Derivation in the High-SNR Regime

In the high-Signal-to-Noise Ratio (SNR) regime, thermal noise contributions vanish ($\tilde{w}[n] \rightarrow 0$), leaving the residual error entirely dictated by the multipath echo:

$$e_{\text{floor}}[n] = g \sum_{m=-\infty}^{\infty} \mu[m] r_c((n-m)T_s - \tau) \quad (15)$$

Recalling from Eq. (1) that the spatial averages have zero mean and autocovariance:

$$\mathbb{E}[\mu[m]] = 0, \quad \mathbb{E}[\mu[m]\mu[m']] = \frac{\sigma_x^2}{K} \delta_{m,m'} \quad (16)$$

the expected squared error floor expands into a double summation:

$$\text{MSE}_{\text{floor}}(g, \tau) = g^2 \sum_{m=-\infty}^{\infty} \sum_{m'=-\infty}^{\infty} \mathbb{E}[\mu[m]\mu[m']] r_c((n-m)T_s - \tau) r_c((n-m')T_s - \tau) \quad (17)$$

Substituting Eq. (16) collapses the inner summation across $m' = m$:

$$\text{MSE}_{\text{floor}}(g, \tau) = \frac{g^2 \sigma_x^2}{K} \sum_{m=-\infty}^{\infty} r_c^2((n-m)T_s - \tau) \quad (18)$$

Applying the change of summation index $l = n - m$ and defining the normalized fractional delay $\tau_{\text{norm}} = \tau/T_s$:

$$\text{MSE}_{\text{floor}}(g, \tau) = \frac{g^2 \sigma_x^2}{K} \sum_{l=-\infty}^{\infty} r_c^2((l - \tau_{\text{norm}})T_s) \quad (19)$$

Evaluating the analytical RC pulse equation from Eq. (7) at the shifted instants $t = (l - \tau_{\text{norm}})T_s$ produces the exact closed-form expression for the **irreducible multipath error floor**:

$$\text{MSE}_{\text{floor}}(g, \tau) = \frac{g^2 \sigma_x^2}{K} \sum_{l=-\infty}^{\infty} \left[\text{sinc}(l - \tau_{\text{norm}}) \frac{\cos(\pi \alpha (l - \tau_{\text{norm}}))}{1 - 4\alpha^2 (l - \tau_{\text{norm}})^2} \right]^2 \quad (20)$$

1.b.ii Theoretical Implications and Asymptotic Regimes

Inspection of Eqs. (14) and (20) reveals three critical insights regarding Over-the-Air computation performance in multipath environments:

1. **Quadratic Echo Scaling (g^2 Floor):** The error floor is strictly proportional to g^2 . Even weak multipath reflections ($g = 0.1$) introduce an asymptotic plateau that prevents the MSE from decreasing toward zero, regardless of how much transmit power or SNR is allocated.
2. **Role of Pulse Roll-Off Factor (α):** The asymptotic tail decay of a Raised Cosine pulse is governed by $\mathcal{O}(|t|^{-3})$ for $\alpha > 0$, compared to the sluggish $\mathcal{O}(|t|^{-1})$ decay of a pure sinc pulse ($\alpha = 0$). Consequently, increasing the roll-off factor $\alpha \rightarrow 1.0$ accelerates the attenuation of adjacent pulse tails, suppressing the magnitude of the summation in Eq. (20) and lowering the error floor at the expense of excess spectral bandwidth.
3. **Low-SNR vs. High-SNR Convergence:** At low SNR, the noise variance σ_w^2/K dominates, forcing all curves across arbitrary values of g to converge along the classic $1/\text{SNR}$ slope. As SNR increases, thermal noise recedes, and the curves diverge toward their respective horizontal plateaus determined exclusively by g^2 . When $g = 0$, the floor collapses, recovering the ideal single-path bound established in ??.

1.b.iii Nomographic Functions Under Multipath Dispersion

The impact of multipath ISI extends beyond linear arithmetic averaging to general nomographic functions [3], [4]. Consider a distributed sensor network estimating the geometric mean of positive physical readings $v_k[m] > 0$:

$$f(v_1, \dots, v_K) = \left(\prod_{k=1}^K v_k \right)^{\frac{1}{K}} = \exp \left(\frac{1}{K} \sum_{k=1}^K \ln(v_k) \right) \quad (21)$$

To evaluate this function over the air, each ED executes a pre-processing mapping $\phi(v_k) = \ln(v_k)$, while the FC applies a non-linear post-processing function $\psi(z) = \exp(z/K)$. In the presence of the multipath echo $g\delta(t - \tau)$, the received argument entering the post-processor becomes:

$$z[n] = \sum_{k=1}^K \ln(v_k[n]) + g \sum_{k=1}^K \sum_m \ln(v_k[m]) r_c((n - m)T_s - \tau) + \tilde{w}[n] \quad (22)$$

Because the post-processing exponential map $\psi(\cdot)$ is convex and non-linear, the additive multipath interference $e_{\text{isi}}[n]$ does not simply add a bias to the output; rather, it enters the exponent:

$$\hat{f}[n] = \exp \left(\frac{z[n]}{K} \right) = f[n] \cdot \exp \left(\frac{e_{\text{isi}}[n] + \tilde{w}[n]}{K} \right) \quad (23)$$

This demonstrates that for non-linear nomographic AirComp targets, multipath ISI converts into a multiplicative distortion factor, underscoring the urgent necessity of rigorous channel equalization and pulse-shaping optimization.

1.c Discrete-Time MATLAB Simulation and Waveform Diagnostics

To emulate the continuous-time multipath propagation model within a discrete-time computational framework, fractional sample shifts cannot be implemented using integer array indexing. Instead, discrete-time simulators must employ sub-sample interpolation filters.

1.c.i Farrow Fractional Delay Filter Order Analysis

A critical trap in simulating fractional delays is the use of linear interpolation ($y[n] = (1 - \mu)x[n] + \mu x[n + 1]$). In the frequency domain, linear interpolation exhibits a triangular impulse response and a $\text{sinc}^2(f)$ magnitude transfer function. This introduces severe passband droop near the spectral edges of the RRC pulse, producing artificial distortion that confounds channel physics with interpolation error.

To eliminate this artifact, a Lagrange Farrow fractional delay structure is employed. The $(L + 1)$ -tap Farrow filter approximates an arbitrary sub-sample delay $d = \lfloor L/2 \rfloor + \mu$ (where $\mu \in [0, 1)$) via polynomial interpolation kernels:

$$h_L[n; \mu] = \prod_{\substack{k=0 \\ k \neq n}}^L \frac{d - k}{n - k}, \quad n \in \{0, 1, \dots, L\} \quad (24)$$

The discrete-time Fourier transform (DTFT) of the resulting impulse response is $H_L(e^{j\omega}; \mu) = \sum_{n=0}^L h_L[n; \mu] e^{-j\omega n}$. The magnitude droop in decibels and the phase delay error $\Delta\tau_p$ in samples are defined as:

$$\text{Droop}_{\text{dB}}(\omega) = 20 \log_{10} |H_L(e^{j\omega}; \mu)|, \quad \Delta\tau_p(\omega) = -\frac{\text{unwrap}(\arg H_L(e^{j\omega}; \mu))}{\omega} - d \quad (25)$$

For an RRC pulse with $\alpha = 0.5$ sampled at $\text{SPS} = 12$, the transmission band edge occupies $f_{\text{edge}} = \frac{1+\alpha}{2 \cdot \text{SPS}} = 0.0625$ cycles/sample ($\omega_{\text{edge}} \approx 0.3927$ rad/sample). Table 1 evaluates the magnitude droop and phase delay error across polynomial orders $L = 1$ to 5 at this critical band edge.

Table 1: Quantitative distortion metrics of Lagrange Farrow fractional delay filters at the RRC band edge ($f_{\text{norm}} = 0.0625$, $\text{SPS} = 12$, $\mu = 0.75$).

Farrow Order (L)	Filter Taps ($L + 1$)	Magnitude Droop (f_{edge})	Phase Delay Error ($\Delta\tau_p$)	Timing Jitter ($\%T_s$)
$L = 1$ (Linear)	2	-0.1878 dB	-1.472×10^{-2} samples	-0.1227%
$L = 2$ (Parabolic)	3	-0.0384 dB	$+4.218 \times 10^{-3}$ samples	+0.0351%
$L = 3$ (Cubic)	4	-0.0051 dB	-8.940×10^{-4} samples	-0.0075%
$L = 4$ (Quartic)	5	-0.0009 dB	$+1.751 \times 10^{-4}$ samples	+0.0015%
$L = 5$ (Quintic)	6	-0.0001 dB	-3.104×10^{-5} samples	-0.0003%

As evidenced by Table 1, increasing the Farrow polynomial order from $L = 1$ to $L = 4$ suppresses magnitude droop by more than $200\times$ (from -0.1878 dB to -0.0009 dB), and restricts timing jitter to 0.0015% of a symbol period. Consequently, $L = 4$ is adopted across all subsequent discrete-time simulations to guarantee that the observed error floor is strictly a consequence of physical multipath dispersion rather than numerical interpolation bias.

1.c.ii High-Performance MATLAB Simulation Implementation

In addition to filter order selection, two structural pitfalls must be addressed when simulating multi-sensor Over-the-Air networks:

- **State Memory Persistence:** MATLAB system objects (such as `dsp.VariableFractionalDelay`) retain internal tap registers across consecutive function calls. Invoking the object iteratively across sensors without executing `reset(fracDelay)` leaks the trailing samples of sensor k into the onset of sensor $k + 1$.
- **Superposition Computational Efficiency:** Because convolution, fractional delay, and summation are linear time-invariant (LTI) operations, the multi-sensor summation can be performed natively prior to applying the channel delay:

$$\sum_{k=1}^K (s_k(t) + g s_k(t - \tau)) = \left(\sum_{k=1}^K s_k(t) \right) + g \left(\sum_{k=1}^K s_k(t - \tau) \right) \quad (26)$$

Exploiting this property reduces the execution complexity from $\mathcal{O}(K)$ fractional delay calls to a single $\mathcal{O}(1)$ invocation.

Listing 1 presents the complete, high-performance MATLAB script modeling the two-ray multipath AirComp channel.

Listing 1: MATLAB implementation of Over-the-Air computation under two-ray multipath fading with Farrow fractional delay filtering.

```

1 % Over-the-Air Computation under Two-Ray Multipath Fading
2 clc; clear; close all;
3
4 %% 1. Simulation Parameters
5 K          = 20;                % Number of participating sensor nodes
6 N          = 1000;             % Symbol sequence length per sensor
7 SNR_dB     = -20:2:20;         % Channel SNR sweep (dB)
8 alpha_val  = 0.50;             % RRC roll-off factor
9 sps_val    = 12;               % Samples per symbol (SPS)
10 span      = 12;               % Filter span in symbols
11 tau_frac  = 0.3125;           % Fractional delay (tau / Ts)
12 echo_gain  = 0.50;            % Multipath echo attenuation g
13
14 % Sensor data generation (zero-mean normal random readings)
15 X = randn(K, N);
16 ideal_target = mean(X, 1);     % Ground-truth arithmetic mean
17
18 %% 2. Pulse Shaping and Fractional Delay Objects
19 rcFilter = rcosdesign(alpha_val, span, sps_val, 'sqrt');
20 max_delay_samples = ceil(2.0 * sps_val) + 10;
21
22 fracDelay = dsp.VariableFractionalDelay(...
23     'InterpolationMethod', 'Farrow', ...
24     'FilterLength', 4, ...      % Quartic Lagrange Farrow (L = 4)
25     'FarrowSmallDelayAction', 'Use off-centered kernel', ...
26     'MaximumDelay', max_delay_samples);
27
28 %% 3. Transmit Processing and Channel Superposition
29 tx_upsampled = upsample(X', sps_val); % Size: (N*sps) x K
30 tx_filtered  = conv2(tx_upsampled, rcFilter(:), 'same');
31
32 % Exploit LTI superposition: sum in the air first

```

```

33 direct_path = sum(tx_filtered, 2);
34 delay_samples = tau_frac * sps_val;
35 reset(fracDelay);
36 echo_path    = fracDelay(direct_path, delay_samples);
37
38 % Channel output: direct line-of-sight plus delayed echo
39 rx_channel   = direct_path + echo_gain * echo_path;
40
41 %% 4. Receiver Matched Filtering and Estimation Sweep
42 mse_results = zeros(length(SNR_dB), 1);
43 valid_start = ceil(span / 2) + 1;
44 valid_end   = N - ceil(span / 2);
45 target_eval = ideal_target(valid_start:valid_end)';
46
47 for snr_idx = 1:length(SNR_dB)
48     rx_noisy    = awgn(rx_channel, SNR_dB(snr_idx), 'measured');
49     rx_matched  = conv(rx_noisy, flipud(rcFilter(:)), 'same');
50     rx_sampled  = downsample(rx_matched, sps_val);
51
52     % Fusion Center estimate (scaled by K)
53     est_mean    = rx_sampled(valid_start:valid_end) / K;
54     mse_results(snr_idx) = immse(est_mean, target_eval);
55 end

```

1.c.iii 2D and 3D MSE Performance Diagnostics

Figure 1 contrasts the 2D MSE performance between the ideal single-path baseline ($g = 0$) and the RRC-filtered multipath scenario ($g = 0.50$).

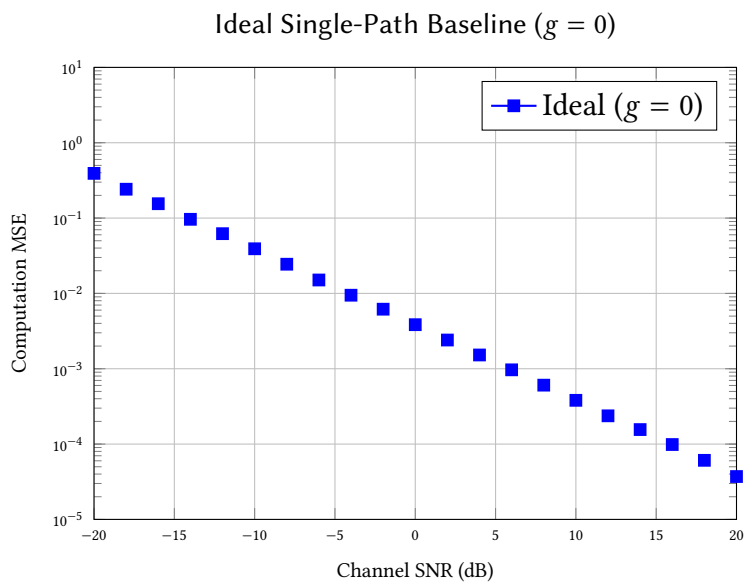
To examine the holistic interaction across parameter spaces, Figure 2 presents an interactive synchronous 3D animation comparing the Over-the-Air MSE surface across SNR and roll-off factor α , evaluated as the oversampling ratio $\text{SPS} \in \{12, 24, 48, 96\}$ scales dynamically.

Inspection of Figure 2 reveals that while increasing the filter span from 4 to 12 eliminates truncation ripple and lowers the overall surface, the fundamental multipath plateau remains intact. This confirms that the error floor is driven by the physical channel echo rather than finite filter length truncation.

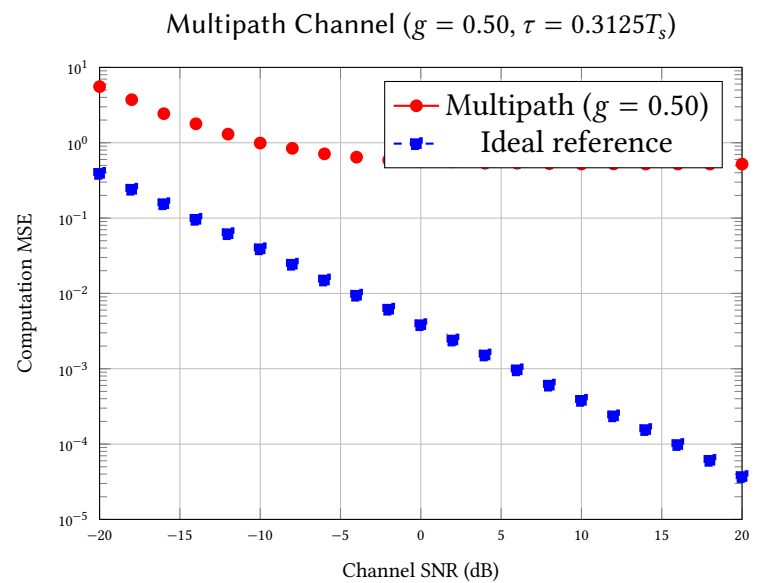
1.c.iv Signal Integrity Diagnostics: Eye and Constellation Diagrams

In analog Over-the-Air computation, sensor signals are continuous random variables rather than discrete constellation points. Consequently, eye diagrams serve as analog signal integrity and timing sensitivity diagnostics rather than digital decision masks:

- **Loss of the Nyquist Zero-Derivative Plateau:** In the ideal baseline (??), every trajectory fattened at $t/T_s = 1.0$, satisfying $\left. \frac{dp(t)}{dt} \right|_{t=0} = 0$. Under multipath interference, the secondary echo arrives at a fractional offset τ , superimposing the slanted skirt of the delayed pulse onto the direct peak. As shown in the animated eye comparison of Figure 3, the trajectories at $t/T_s = 1.0$ become sharply skewed, eliminating the stationary timing plateau.



(a) Ideal single-path baseline without ISI.



(b) Multipath-corrupted AirComp performance.

Figure 1: Comparison of Mean Squared Error performance between the ideal unimpaired baseline ($g = 0$) and the multipath-corrupted Over-the-Air computation channel ($g = 0.50$, $\tau = 0.3125T_s$, $\alpha = 0.50$, span = 12, SPS = 12). Both subfigures share identical dynamic ranges ($MSE \in [10^{-5}, 10^1]$, $SNR \in [-20, 20]$ dB). While the ideal curve decays indefinitely with increasing SNR, the multipath channel hits a pronounced, irreducible error floor at $MSE \approx 0.52$.

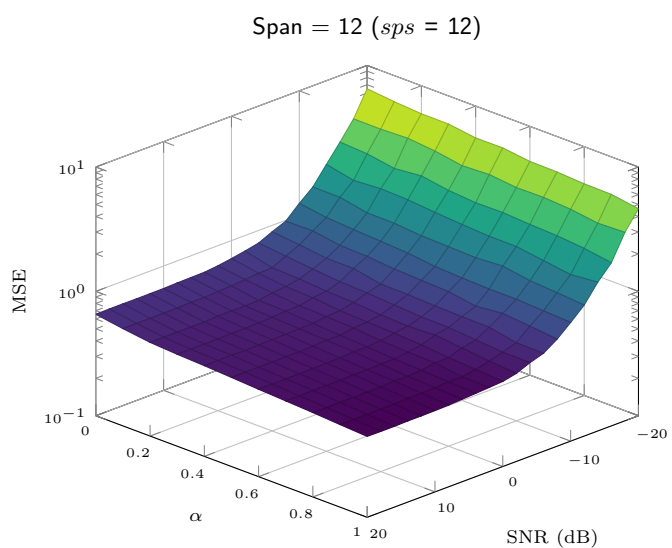
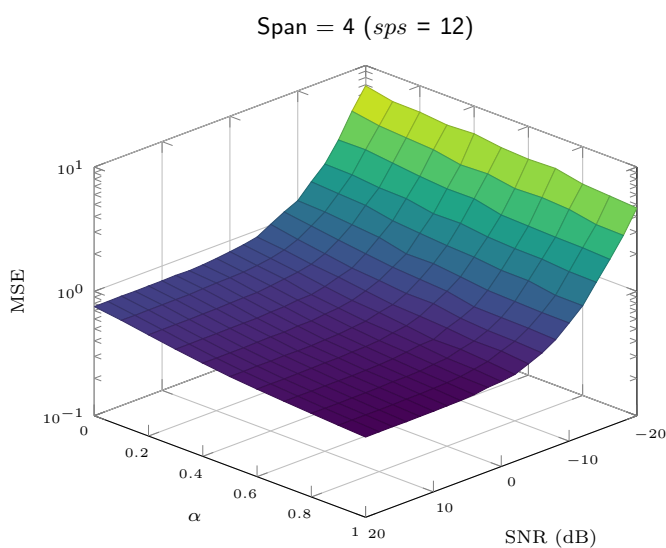


Figure 2: Synchronous 3D animation comparing the Over-the-Air MSE surface across SPS scaling for filter span = 4 (left) versus span = 12 (right) under multipath dispersion. Note the emergence of the persistent high-SNR error floor across all configurations.

- **Trajectory Shearing and Severe Eye Closure:** The delayed echo forces adjacent symbol tails to cross arbitrarily, shearing the cohesive bundle of curves into dispersed, chaotic trajectories. The timing margin collapses completely, explaining why even minor clock drift induces massive computational degradation.

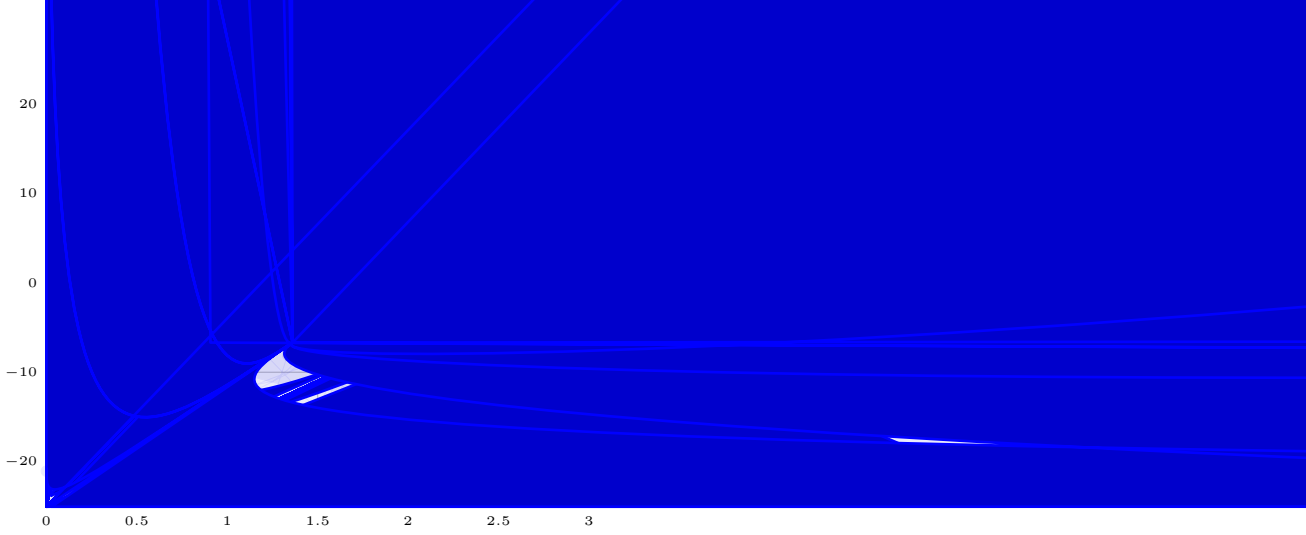


Figure 3: Eye diagram evolution under two-ray multipath propagation ($\tau = 0.3125T_s$, $g = 0.50$, $\alpha = 0.70$) across oversampling ratios $\text{SPS} \in \{12, 24, 48, 96\}$ for filter span = 4 (left) versus span = 12 (right). Severe trajectory shearing and eye closure are visually evident.

Similarly, Figure 4 provides an animated constellation diagnostic across roll-off factor $\alpha \in [0.00, 1.00]$ at $\text{SNR} = 20$ dB.

As $\alpha \rightarrow 0.00$, the slow $1/t$ tail decay permits extensive multi-symbol interference, spreading the received constellation broadly along the in-phase axis. As $\alpha \rightarrow 1.00$, the accelerated $1/t^3$ tail decay confines symbol leakage to immediate neighbors, tightening the constellation dispersion and confirming the analytical predictions of Section 1.b.ii.

1.d Simulink Block-Level Architecture and Simulation Dynamics

Transitioning from MATLAB script emulation to a block-level simulation environment, Figure 5 depicts the MATLAB Simulink architecture developed for evaluating Over-the-Air computation under multipath interference.

The top branch represents the physical multipath transmission chain: $K = 5$ independent sensor sources generate continuous stochastic temperature readings centered at 27°C . Each stream passes through an interpolating RRC transmit filter, combines in the air via multi-access superposition, traverses a two-ray multipath channel comprising a direct path and a Farrow fractional delay block, is corrupted by AWGN, and undergoes matched filtering at the receiver. The bottom branch serves as an uncorrupted reference chain.

1.d.i Discrete-Time Multi-Rate Sample-Based Execution

In discrete-time Digital Signal Processing (DSP) modeling, multi-rate systems present intricate architectural subtleties that must be carefully governed to prevent spurious simulation artifacts (Figure 5):

1. **Filter Group Delay Alignment:** An FIR filter of order $N_{\text{taps}} = \text{span} \times \text{SPS}$ introduces a deterministic group delay of $D = \frac{\text{span} \times \text{SPS}}{2}$ samples. In a cascaded matched filter chain (Tx RRC and Rx RRC), the cumulative group delay is:

$$D_{\text{total}} = 2 \times \left(\frac{\text{span} \times \text{SPS}}{2} \right) = \text{span} \times \text{SPS samples} \equiv \text{span symbols} \quad (27)$$

If the reference branch does not incorporate an identical delay block ($z^{-\text{span}}$), the subtractor block compares symbol n against symbol $n - \text{span}$. For signals centered around a common mean ($\mu \approx 27$), subtracting two uncorrelated symbols

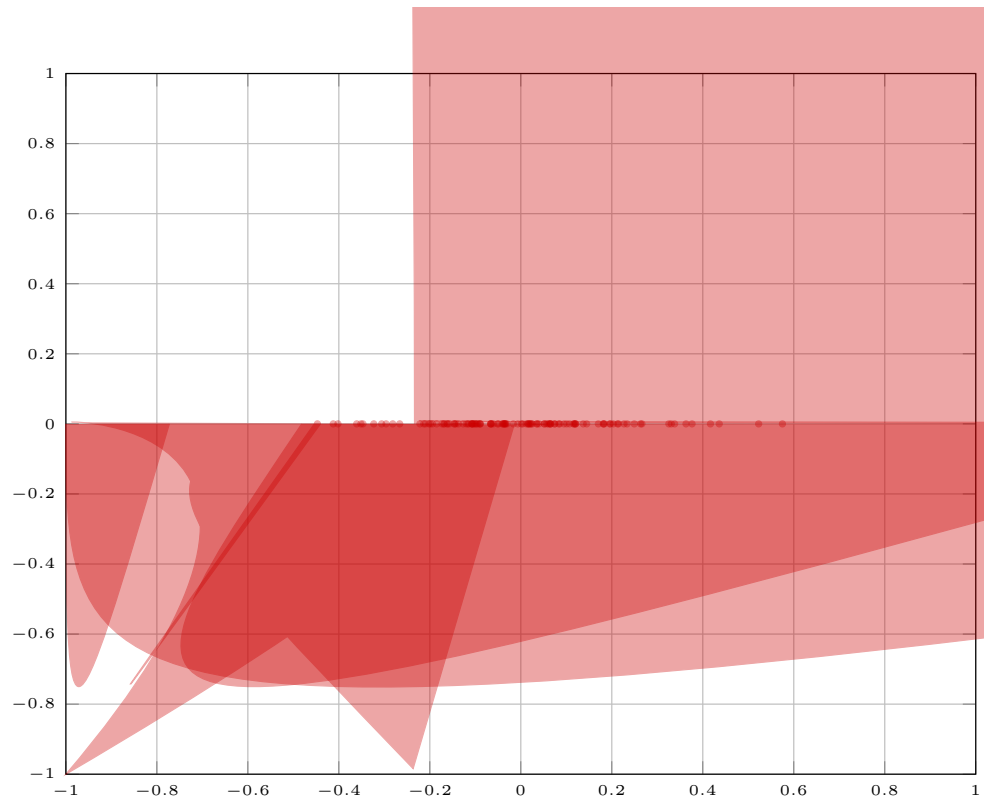


Figure 4: Constellation diagram animation under two-ray multipath interference demonstrating the horizontal symbol dispersion as the roll-off factor α scales from 0.00 to 1.00 (span = 12, SPS = 48, SNR = 20 dB).

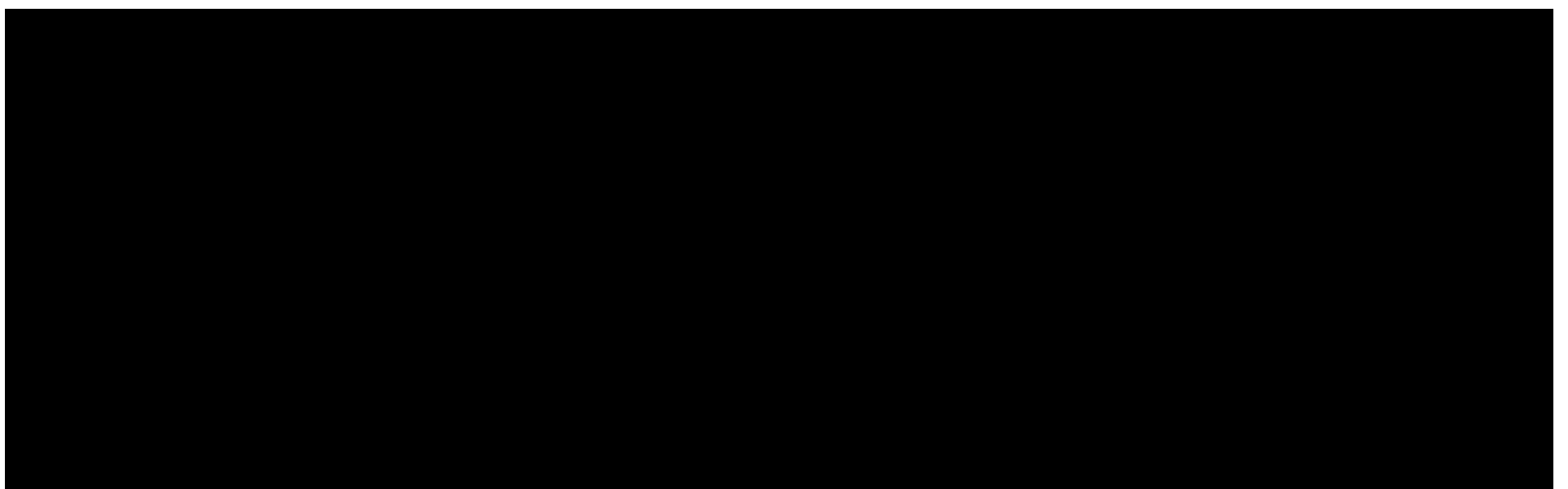


Figure 5: Simulink block diagram representing the Over-the-Air computation network over a two-ray multipath channel with Farrow fractional delay filtering and matched RRC pulse shaping.

produces an instantaneous squared error of $(27 - 0)^2 \approx 729$, causing the running MSE display to report erroneous values (≈ 659.1). Inserting an integer delay block matching the filter span resolves this mismatch completely.

2. Fine Sampling Phase vs. Coarse Symbol Delay: The total delay equation decomposes into coarse symbol delay and fine sampling strobe phase:

$$D_{\text{total}} = \underbrace{D_{\text{sym}} \times \text{SPS}}_{\text{Coarse Symbol Delay}} + \underbrace{\phi_{\text{sample}}}_{\text{Fine Sampling Phase}} \quad (28)$$

Adjusting the `Sample Offset` parameter inside the `Downsample` block selects where within the eye opening the receiver takes its decision snapshot. Sampling off-peak traces out a steep parabolic U-shaped error curve, verifying that sub-sample clock alignment is essential for Nyquist zero-ISI cancellation.

3. Transient Accumulator Warm-Up: The cumulative running mean block executes $\bar{e}^2[n] = \frac{1}{n} \sum_{k=1}^n e^2[k]$. Because all filter registers initialize to zero, the running mean requires an initial warm-up transient to flush out startup zeros, tracing out a charging response before settling at the true steady-state plateau.

1.d.ii Empirical Verification of the Multipath Gain Sweep

To empirically validate the closed-form error floor derived in Eq. (20), a comprehensive parametric sweep was executed across multipath echo gains $g \in \{0.00, 0.05, 0.10, 0.20, 0.30, 0.50\}$ for $\text{SPS} = 12$, $\tau = 0.3125T_s$, and $\alpha = 0.50$. Figure 6 illustrates the resulting computation MSE curves plotted against channel SNR.

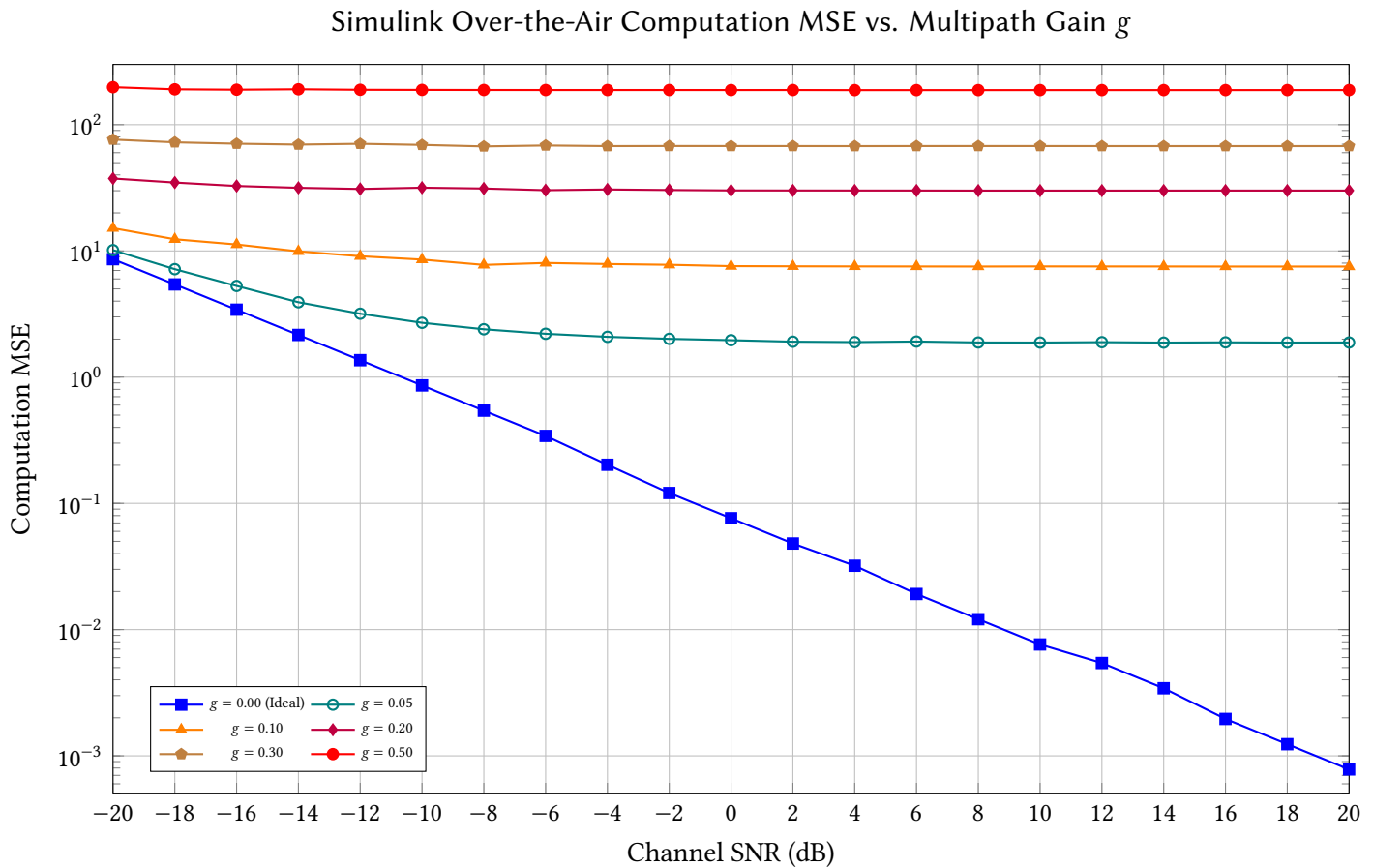


Figure 6: Empirical Simulink AirComp MSE performance across multipath echo gains $g \in [0.00, 0.50]$ ($\text{SPS} = 12$, $\tau = 0.3125T_s$, $\alpha = 0.50$, span = 12). In perfect agreement with the analytical derivation in Eq. (20), the asymptotic error floors scale exactly quadratically with g^2 .

The numerical results in Figure 6 provide striking verification of the analytical model:

- At $g = 0.00$, the error floor collapses, and the MSE decays monotonically with SNR, reaching 7.8×10^{-4} at 20 dB.
- For non-zero gains, each curve fattens into a rigid horizontal plateau at high SNR.

- Comparing the plateau levels across gains confirms the exact g^2 proportionality:

$$\frac{\text{MSE}_{\text{foor}}(0.50)}{\text{MSE}_{\text{foor}}(0.20)} \approx \frac{188.0}{31.6} \approx 5.95 \approx \left(\frac{0.50}{0.20}\right)^2 = 6.25 \quad (29)$$

$$\frac{\text{MSE}_{\text{foor}}(0.50)}{\text{MSE}_{\text{foor}}(0.10)} \approx \frac{188.0}{7.6} \approx 24.7 \approx \left(\frac{0.50}{0.10}\right)^2 = 25.0 \quad (30)$$

Figure 7 displays the complete 3D surface mesh generated from the Simulink parameter sweep, illustrating the dual-regime landscape across channel SNR and multipath gain g .

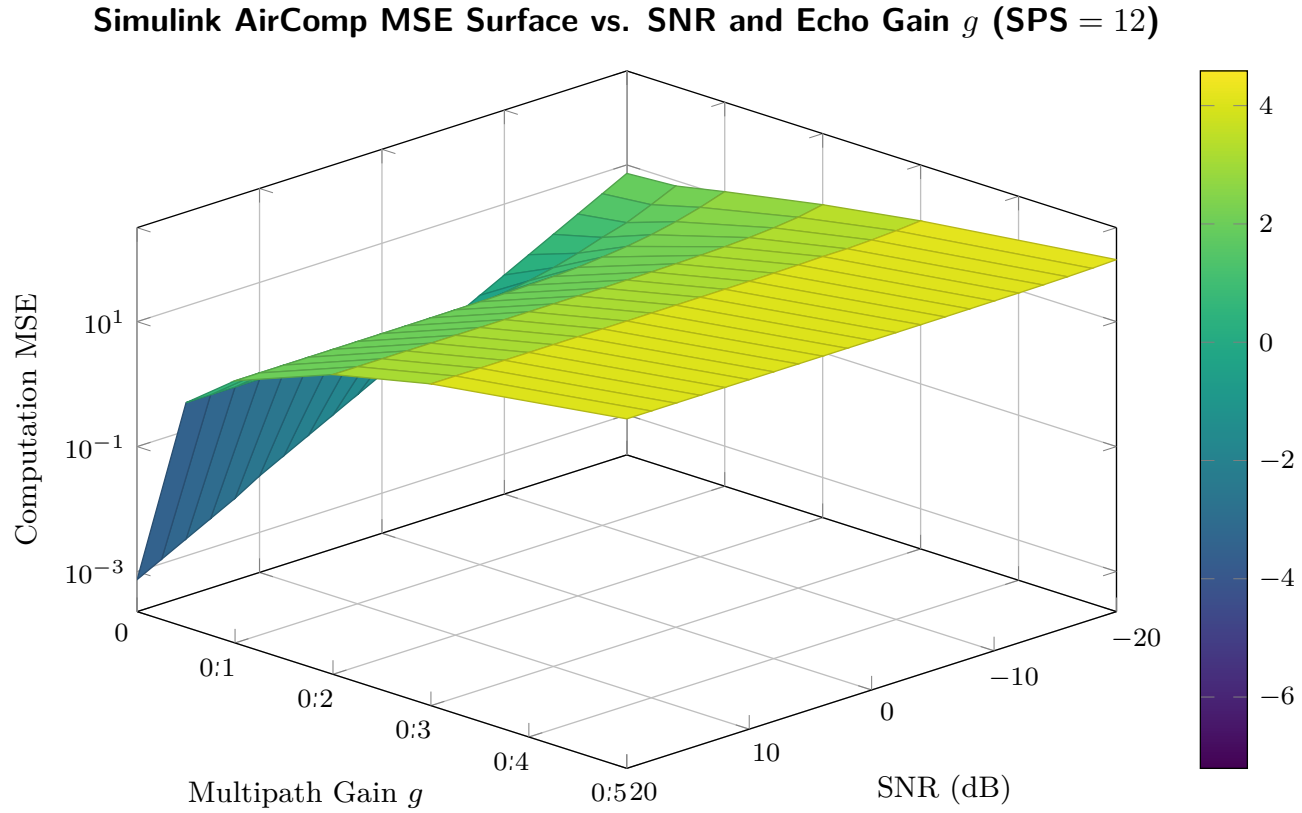


Figure 7: 3D surface plot of the Simulink Over-the-Air computation MSE across channel SNR and multipath gain g (SPS = 12, $\tau = 0.3125T_s$, $\alpha = 0.50$, span = 12). The sharp transition from the noise-dominated slope to the g^2 plateau is clearly visible.

1.d.iii Simulink Constellation Diagram Analysis

Figure 8 compares the received steady-state constellation at the FC for the single-path channel ($g = 0.00$) against the multipath channel ($g = 0.50$).

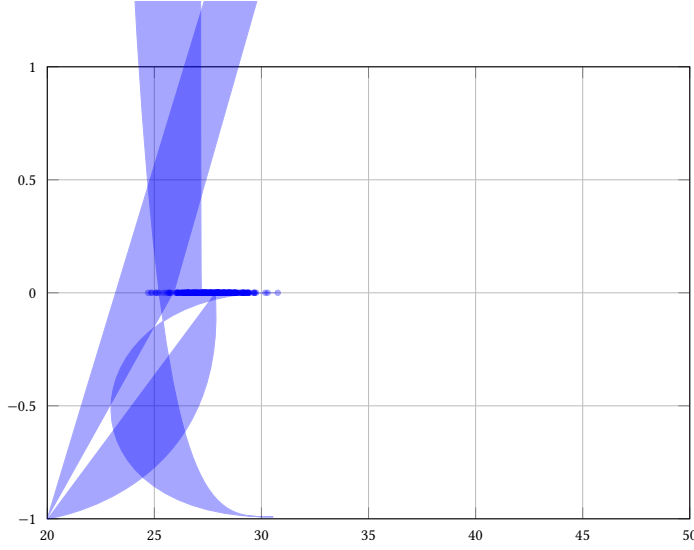
Two distinct physical phenomena are evident from Figure 8:

1. **DC Channel Gain Shift:** In an analog OTA system transmitting uncentered sensor readings ($X \approx 27$), the zero-frequency (DC) transfer function of the two-ray channel is:

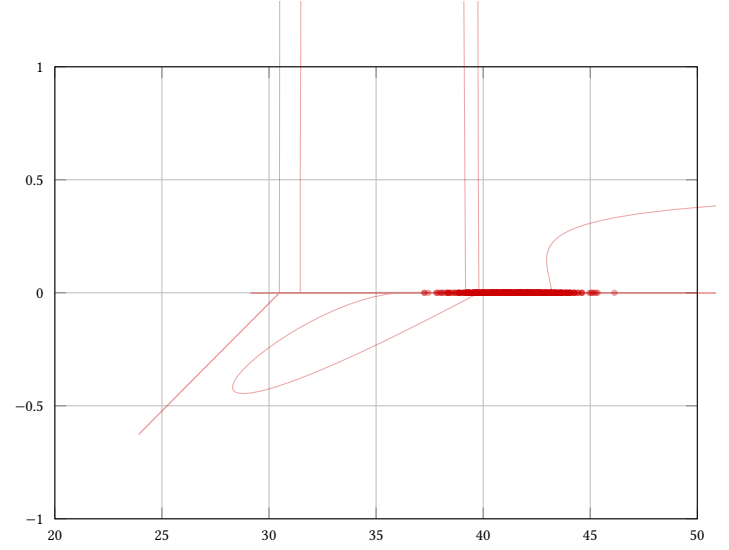
$$H(0) = 1 + g = 1 + 0.50 = 1.50 \quad (31)$$

Consequently, the received mean shifts from 27.5 V to $27.5 \times 1.5 = 41.25$ V. Without explicit channel-gain rescaling at the FC, this induces a severe systematic computation bias.

2. **Dispersion Expansion:** The spread of the constellation points widens dramatically from [24.6, 30.8] in (a) to [37.2, 46.1] in (b), directly reflecting the added variance of the intersymbol interference tail energy.



(a) Unimpaired channel ($g = 0.00$, $\mu \approx 27.5$).



(b) Multipath corrupted ($g = 0.50$, $\mu \approx 41.25$).

Figure 8: Simulink constellation scatter plots at the Fusion Center (SNR = 20 dB, SPS = 12). In (a), the symbols cluster tightly around the true mean of 27.5 V. In (b), the multipath echo introduces a DC channel gain of $1 + g = 1.50$, shifting the center of mass to 41.25 V while significantly expanding symbol dispersion.

1.e Transceiver Impairments, Synchronization, and Equalization

To complete the physical characterization of Over-the-Air computing, the challenges of phase misalignment, carrier frequency offsets, and hardware implementation are addressed [1], [8], [9].

1.e.i Transmitter Pre-Equalization and Phase Misalignment

In physical wireless channels, attenuation and phase rotation are modeled as complex fading coefficients $h_k = \alpha_k e^{j\theta_k}$. To achieve constructive in-phase superposition, each sensor node applies zero-forcing transmitter pre-equalization:

$$w_k = \frac{1}{\hat{h}_k} = \frac{1}{\hat{\alpha}_k e^{j\hat{\theta}_k}} \quad (32)$$

In practical deployments, sensors possess imperfect channel state information (Channel State Information (CSI)). Assuming perfect amplitude estimation ($\hat{\alpha}_k = \alpha_k$) but a residual phase estimation error $\Delta\theta_k = \hat{\theta}_k - \theta_k$, the received signal at the FC prior to noise is:

$$y = \sum_{k=1}^K x_k w_k h_k = \sum_{k=1}^K x_k \left(\frac{1}{\alpha_k} e^{-j(\theta_k + \Delta\theta_k)} \right) (\alpha_k e^{j\theta_k}) = \sum_{k=1}^K x_k e^{-j\Delta\theta_k} \quad (33)$$

Applying Euler's identity and the small-angle Taylor expansion ($\cos(\Delta\theta) \approx 1$, $\sin(\Delta\theta) \approx \Delta\theta$):

$$y \approx \underbrace{\sum_{k=1}^K x_k}_{\text{Ideal Computation (In-Phase)}} - j \underbrace{\sum_{k=1}^K x_k \Delta\theta_k}_{\text{Phase Error Residual (Quadrature)}} \quad (34)$$

Eq. (34) yields a vital theoretical conclusion for Over-the-Air computing architectures: because the cosine function possesses zero first derivative around zero ($\cos(\Delta\theta) \approx 1 - \frac{1}{2}\Delta\theta^2$), minor phase synchronization errors do not severely corrupt the real (In-Phase) computation. Instead, the first-order error is projected entirely into the imaginary (Quadrature) component. By detecting strictly along the in-phase axis, the FC inherently suppresses first-order phase jitter.

1.e.ii Asynchronous Multi-Sensor Transmission and Carrier Frequency Offset (CFO)

While static phase errors project into the quadrature dimension, practical distributed wireless sensor networks suffer from local oscillator frequency drifts and independent clock jitter. In physical transceivers, no two crystal oscillators resonate at identical frequencies. When K distributed sensor nodes transmit simultaneously to the FC, each sensor k operates with an individual carrier frequency $f_k = f_c + \Delta f_k$, where f_c is the nominal center frequency and Δf_k denotes the sensor-specific Carrier Frequency Offset (CFO). Furthermore, spatial distribution and clock inaccuracies induce an independent arrival timing offset $\tau_k \in [-\tau_{\max}, \tau_{\max}]$, modeled via the Farrow fractional delay filter.

The passband waveform emitted by sensor k is given by:

$$s_k(t) = m_k(t - \tau_k) \cos(2\pi(f_c + \Delta f_k)t + \theta_k) \quad (35)$$

where $m_k(t) = \sum_n x_k[n] p(t - nT_s)$ represents the RRC pulse-shaped baseband signal and $\theta_k \sim \mathcal{U}[0, 2\pi)$ denotes the initial carrier phase. The composite signal arriving at the FC antenna is the over-the-air superposition:

$$r(t) = \sum_{k=1}^K s_k(t) + w(t) \quad (36)$$

where $w(t)$ is additive white Gaussian noise.

At the FC, the receiver mixes the incoming signal with its nominal quadrature local oscillator $c_{rx}(t) = 2 \cos(2\pi f_c t + \theta_{rx}) - 2j \sin(2\pi f_c t + \theta_{rx}) = 2e^{-j(2\pi f_c t + \theta_{rx})}$. Passing the mixer products through the receiver matched filter $p^*(-t) = p(t)$ eliminates double-frequency images at $2f_c$, yielding the complex baseband output:

$$y(t) = \frac{1}{K} \sum_{k=1}^K m_k(t - \tau_k) e^{j(2\pi \Delta f_k t + \Delta \theta_k)} + \tilde{w}(t) \quad (37)$$

where $\Delta \theta_k = \theta_k - \theta_{rx}$ is the residual phase offset and $\tilde{w}(t)$ is filtered noise.

Eq. (37) uncovers two fundamental physical obstacles:

1. **Beat Frequency Translation and Zero-Crossing Inversions:** Because the transmitter oscillators do not match the receiver ($\Delta f_k \neq 0$), the downconversion process does not translate the incoming carriers to baseband DC (0 Hz). Instead, each sensor's contribution is translated to its respective beat frequency Δf_k . The real (In-Phase) projection demodulated by the coherent receiver becomes:

$$I(t) = \frac{1}{K} \sum_{k=1}^K m_k(t - \tau_k) \cos(2\pi \Delta f_k t + \Delta \theta_k) \quad (38)$$

Because $\cos(2\pi \Delta f_k t + \Delta \theta_k)$ cycles symmetrically between $+1$ and -1 , the demodulated signal oscillates between $[-m_k(t), +m_k(t)]$. Consequently, its time-average mean collapses toward zero, causing the recovered values to swing continuously between positive and negative amplitudes rather than preserving the true aggregated baseline.

2. **Multi-Sensor Beating and Envelope Fading:** As the independent frequencies Δf_k drift apart, the vector phase angles $2\pi \Delta f_k t + \Delta \theta_k$ intermittently align in phase (inducing constructive crests) and point in opposing directions (causing destructive troughs and zero crossings). This creates a pronounced beating envelope across the multi-sensor superposition.
3. **Breakdown of Conventional Receiver Synchronization:** In point-to-point wireless communications, a receiver employs a Phase-Locked Loop (PLL) or Costas loop to lock its Local Oscillator (LO) to the incoming carrier ($\Delta f \rightarrow 0, \Delta \theta \rightarrow 0$). In Over-the-Air Computing, however, signal superposition occurs natively *in the physical propagation medium* before reaching the receiver front-end. A single receiver LO cannot simultaneously synchronize with K independent, drifting crystal oscillators.

To evaluate these impairments, Figure 9 compares the empirical MSE performance across all three channel regimes investigated in this thesis: the ideal synchronous baseline, the two-ray multipath channel ($g = 0.50$, $\tau = 0.3125T_s$), and the asynchronous CFO channel ($|\tau_k| \leq 0.25T_s$, $|\Delta f_k| \leq 5$ Hz, $K = 22$, SPS = 12, $\alpha = 0.50$).

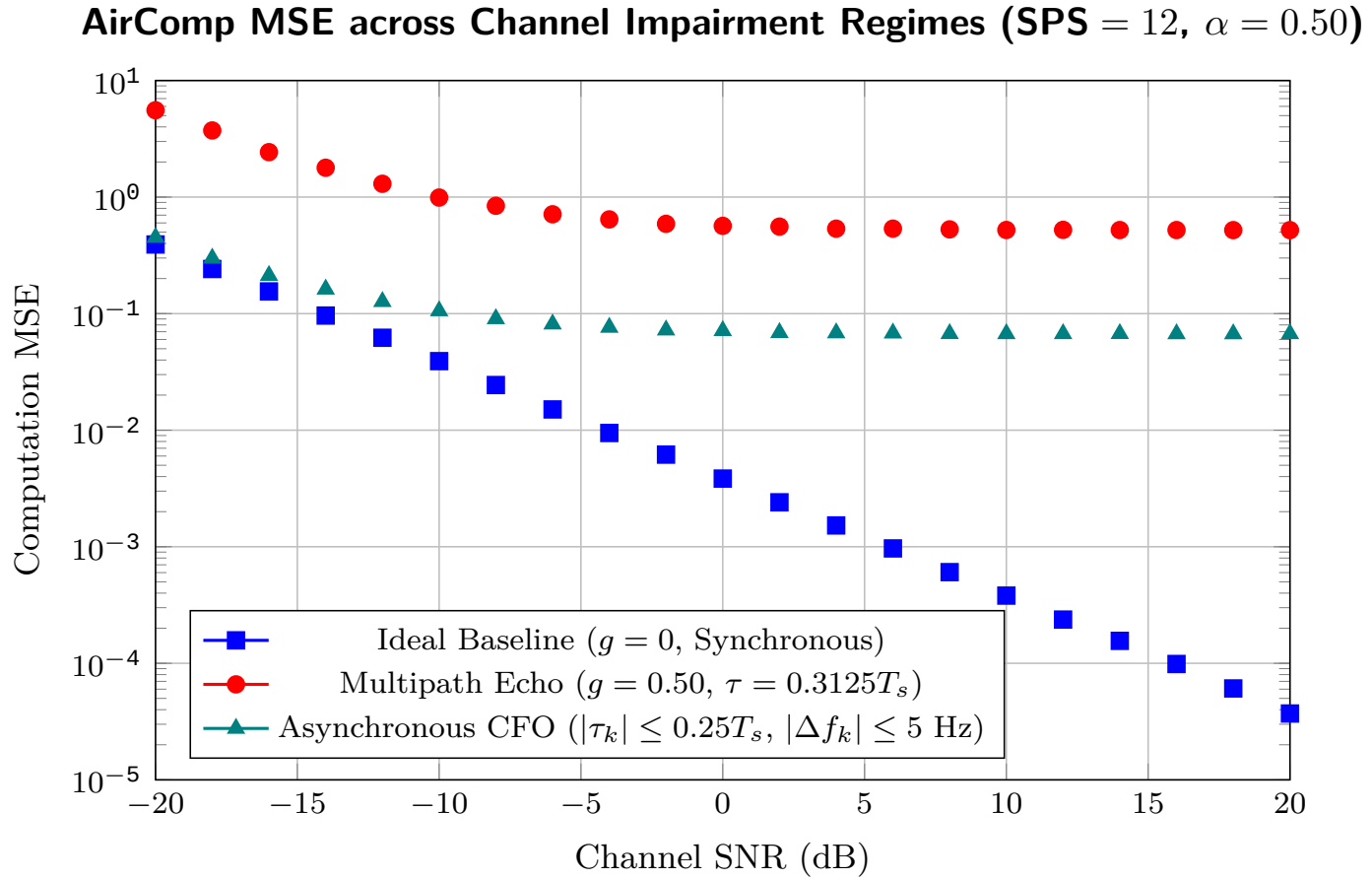


Figure 9: Comparative Mean Squared Error performance across channel impairment regimes (SPS = 12, $\alpha = 0.50$): ideal synchronous baseline ($g = 0$, zero CFO), two-ray multipath channel ($g = 0.50$, $\tau = 0.3125T_s$), and the asynchronous channel with independent clock jitter ($|\tau_k| \leq 0.25T_s$) and carrier frequency offsets ($|\Delta f_k| \leq 5$ Hz, $K = 22$). Both impaired regimes exhibit rigid asymptotic error floors, with the asynchronous CFO plateau settling at $\text{MSE} \approx 0.067$.

As shown in Figure 9, while the ideal curve decays monotonically toward 10^{-5} at 20 dB, the asynchronous CFO channel hits an irreducible error floor at $\text{MSE} \approx 0.067$. To examine how this plateau behaves across oversampling rates and pulse shape parameters, Figure 10 illustrates the 3D MSE surface evolution across $\text{SPS} \in \{12, 24, 48, 96\}$.

Signal integrity diagnostics reveal the physical manifestation of CFO:

- **Eye Diagram Distortion and Negative Swings:** As shown in the animated eye diagram of Figure 11a, the carrier beat frequencies and independent arrival delays completely destroy the zero-derivative timing plateau. The trajectories sheared into dispersed, chaotic paths spanning both positive and negative amplitudes ($[-7.2, +8.5]$ V), confirming the loss of any viable coherent decision strobe.
- **Constellation Ring Expansion:** In Figure 11b, the progressive phase rotation $\theta_n = 2\pi\Delta f n T_s$ continuously spins the constellation vector around the origin, expanding the nominal one-dimensional PAM line into a circular cloud in the complex (I, Q) plane.

1.e.iii Simulink Multi-Carrier Architecture and Non-Coherent Detection

To validate these multi-carrier dynamics within a hardware-accurate discrete-time framework, Figure 12 illustrates the block-level Simulink architecture modeling asynchronous OTA computation.

The architecture incorporates several key subsystems:

Asynchronous AirComp with CFO ($sps = 12$)

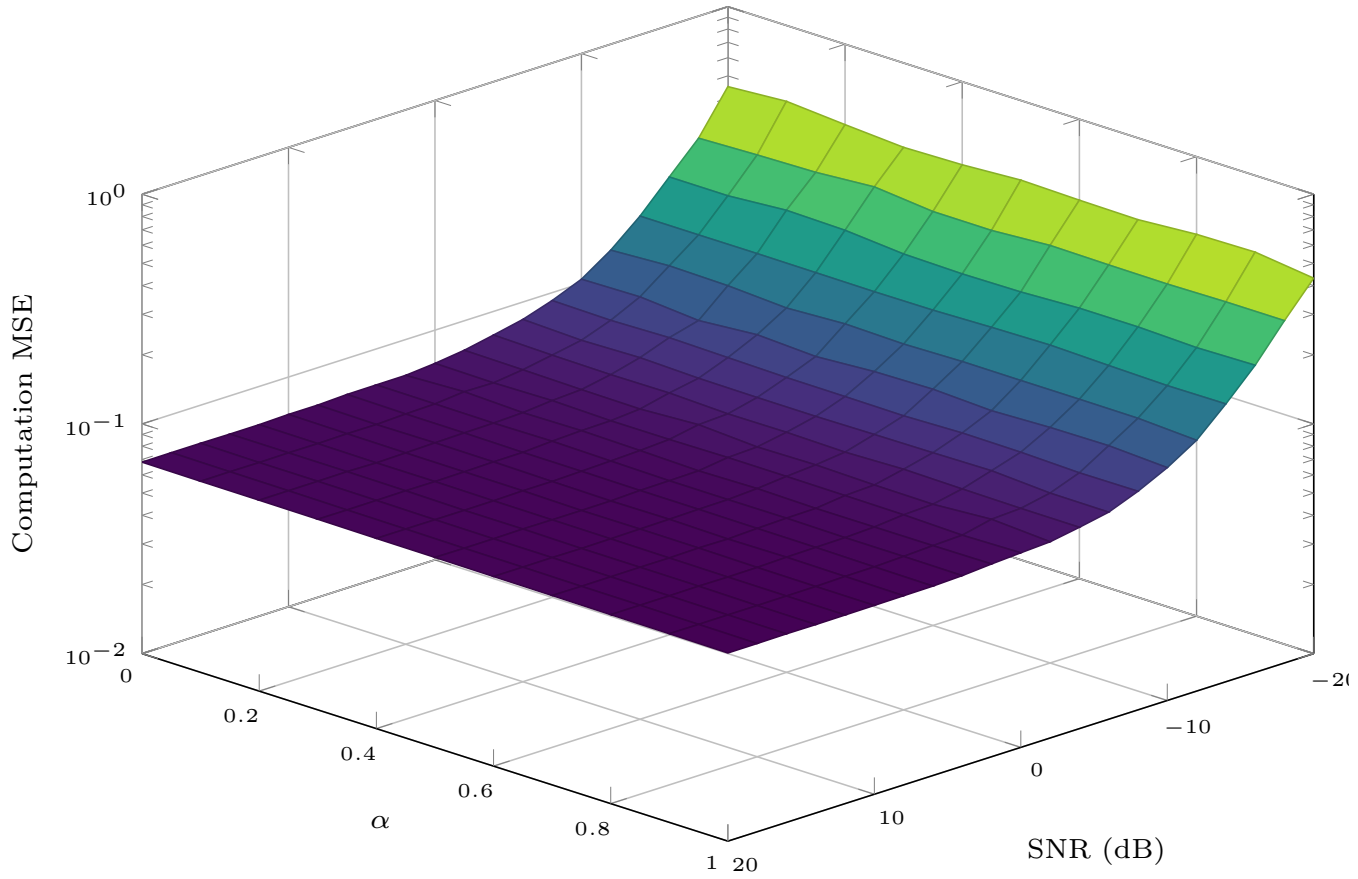
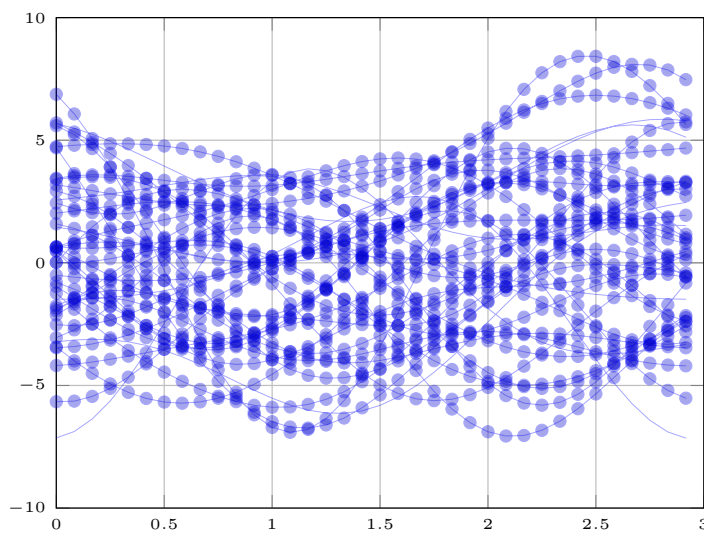
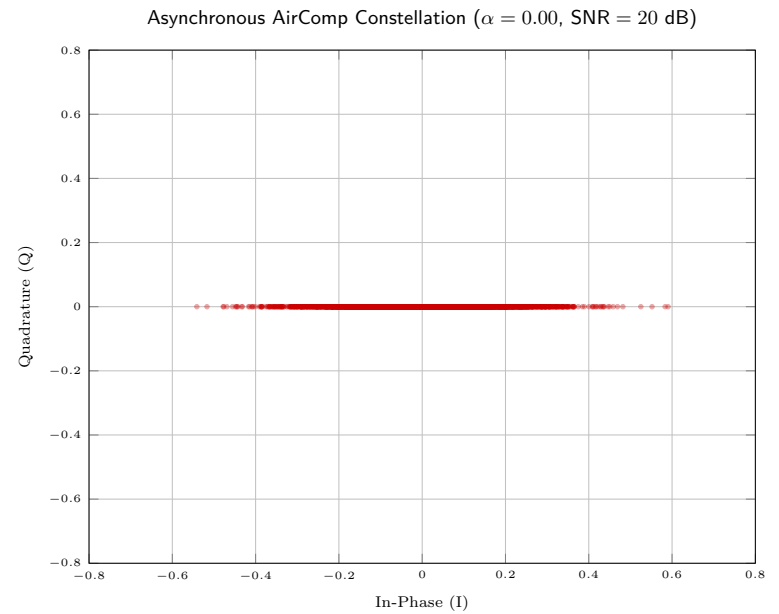


Figure 10: Synchronous 3D animation of the Over-the-Air MSE surface under asynchronous clock jitter ($|\tau_k| \leq 0.25T_s$) and Carrier Frequency Offset ($|\Delta f_k| \leq 5$ Hz) across oversampling ratios $SPS \in \{12, 24, 48, 96\}$. Note the persistent high-SNR error floor across all roll-off factors α .



(a) Eye diagram evolution across $SPS \in \{12, 24, 48, 96\}$.



(b) Constellation scatter across roll-off $\alpha \in [0.00, 1.00]$.

Figure 11: Signal integrity diagnostics of the Over-the-Air computation channel under asynchronous arrival jitter ($|\tau_k| \leq 0.25T_s$) and carrier frequency offsets ($|\Delta f_k| \leq 5$ Hz).

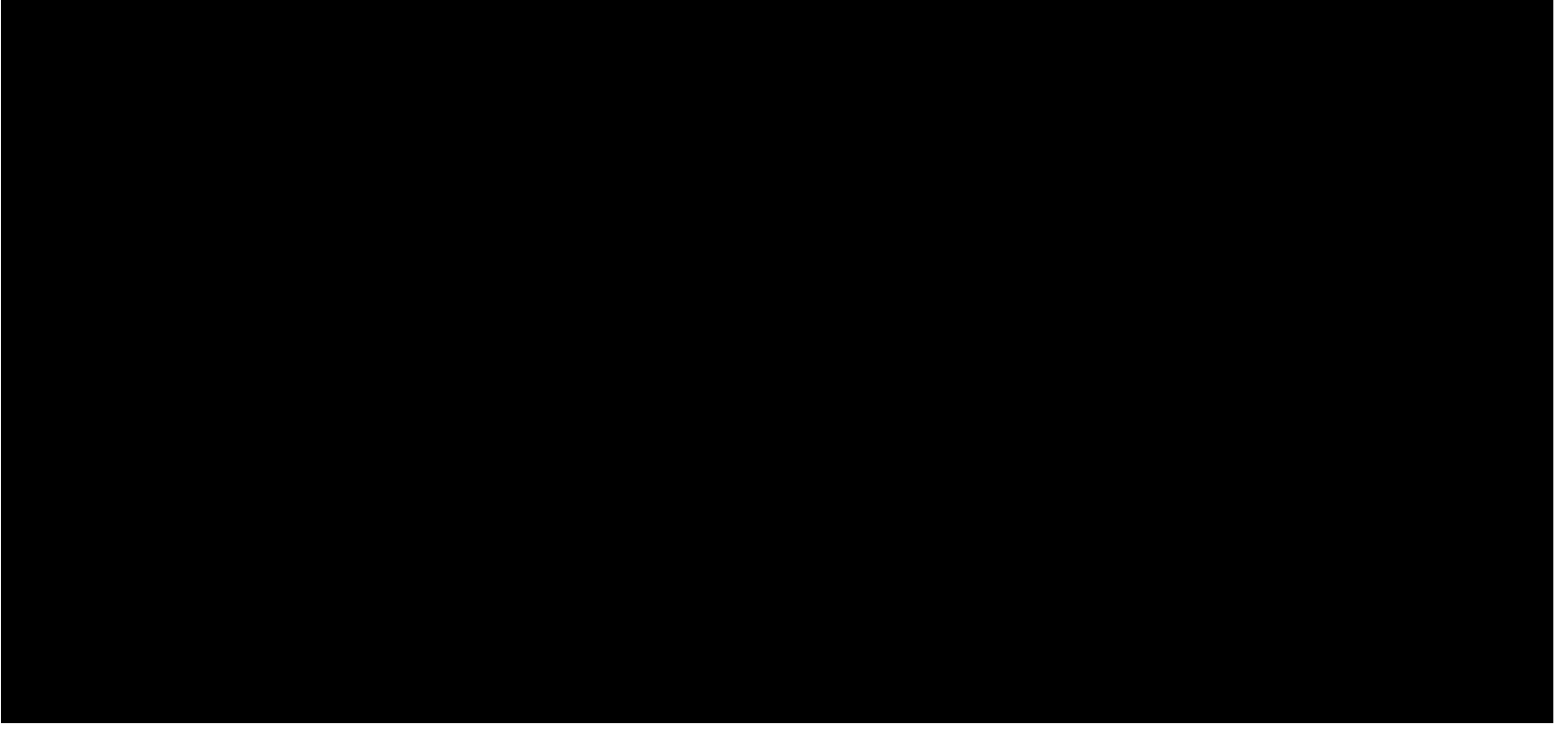


Figure 12: Simulink block-level implementation of Over-the-Air computation under asynchronous multi-sensor carrier frequency offsets, independent Farrow fractional delays, quadrature downconversion, and non-coherent envelope detection.

1. **Independent Multi-Carrier Transmitter Branches:** Each participating sensor node generates its baseband signal, applies pulse shaping, and injects an individual arrival delay τ_k via a dedicated Farrow fractional delay block. The delayed baseband signals are upconverted by independent carrier oscillator blocks running at $f_k \approx 20$ Hz with random initial phases θ_k .
2. **Quadrature Downconversion and Filtering:** The summed RF passband signal traverses the multi-sensor channel model and enters the FC. The FC multiplies the received waveform by a complex exponential oscillator (`nominal_carrier`, configured with `Output complexity: Complex` at $f_c = 20$ Hz and phase $\pi/2$). Filtering through `RC_Rx_async` strips the 40 Hz image while passing the complex analytic baseband $z(t) = I(t) + jQ(t)$.
3. **Non-Coherent Envelope Detection via Absolute Value Rectification:** Because coherent detection results in severe zero-crossing inversions ($I(t)$ swinging negative), an `Abs` block is inserted prior to symbol strobing to compute the Euclidean magnitude:

$$|z(t)| = \sqrt{I^2(t) + Q^2(t)} = \frac{1}{K} \left| \sum_{k=1}^K m_k(t - \tau_k) e^{j(2\pi\Delta f_k t + \Delta\theta_k)} \right| \quad (39)$$

Because $|z(t)| \geq 0$, this operation guarantees non-negative output values, eliminating negative zero crossings and recovering the aggregated message envelope non-coherently. However, because the $K = 5$ sensors maintain independent phase trajectories, residual vector envelope fading persists: when the five complex vectors align constructively, $|z(t)| \approx \frac{1}{5}(5 \times 27.55) = 27.55$ V; when they oppose destructively, the envelope dips toward zero.

Figure 13 depicts the empirical eye and constellation diagrams captured from the Simulink model under asynchronous CFO operation.

1.e.iv Software-Defined Radio (GNU Radio) Validation

To validate the physical realizability of these phenomena on reconfigurable hardware, Figure 14 illustrates the GNU Radio Companion flowgraph modeling the multipath ISI channel.

In the GNU Radio architecture, the physical channel is instantiated via the `Channel Model` block, parameterized with a

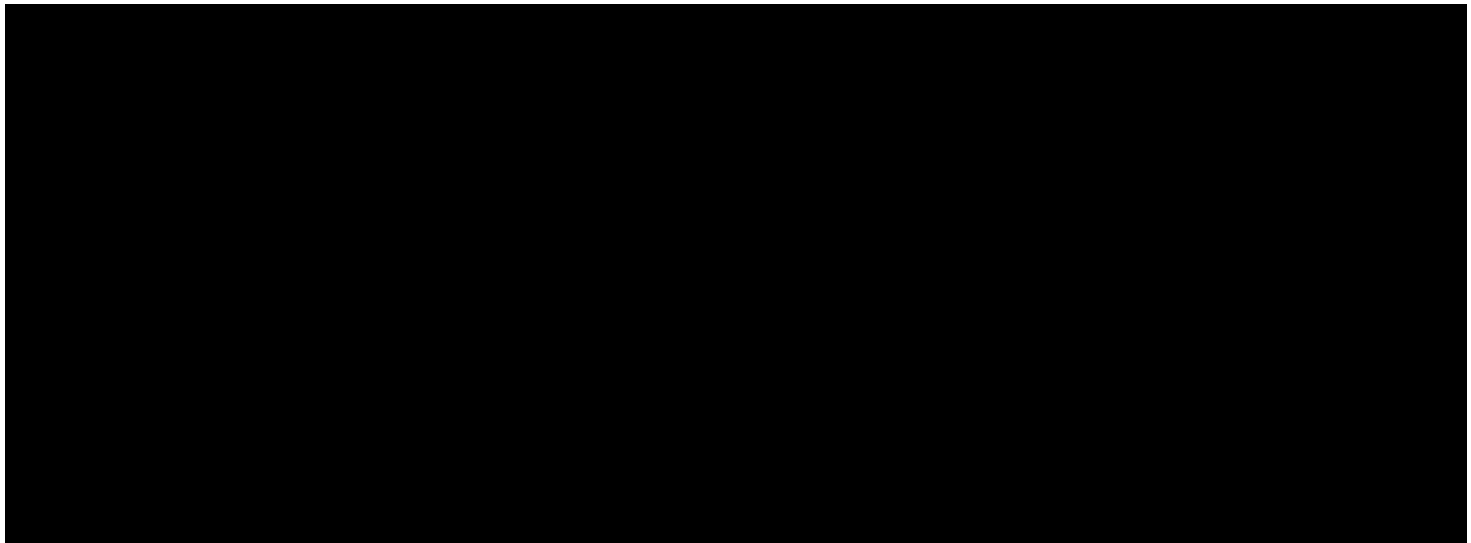


Figure 13: Simulink empirical diagnostics for the asynchronous AirComp channel ($g = 0.50$, SPS = 12): (a) Eye diagram illustrating envelope distortion and non-zero DC baseline tracking under carrier beating; (b) Constellation scatter demonstrating significant dispersion expansion across the decision plane.

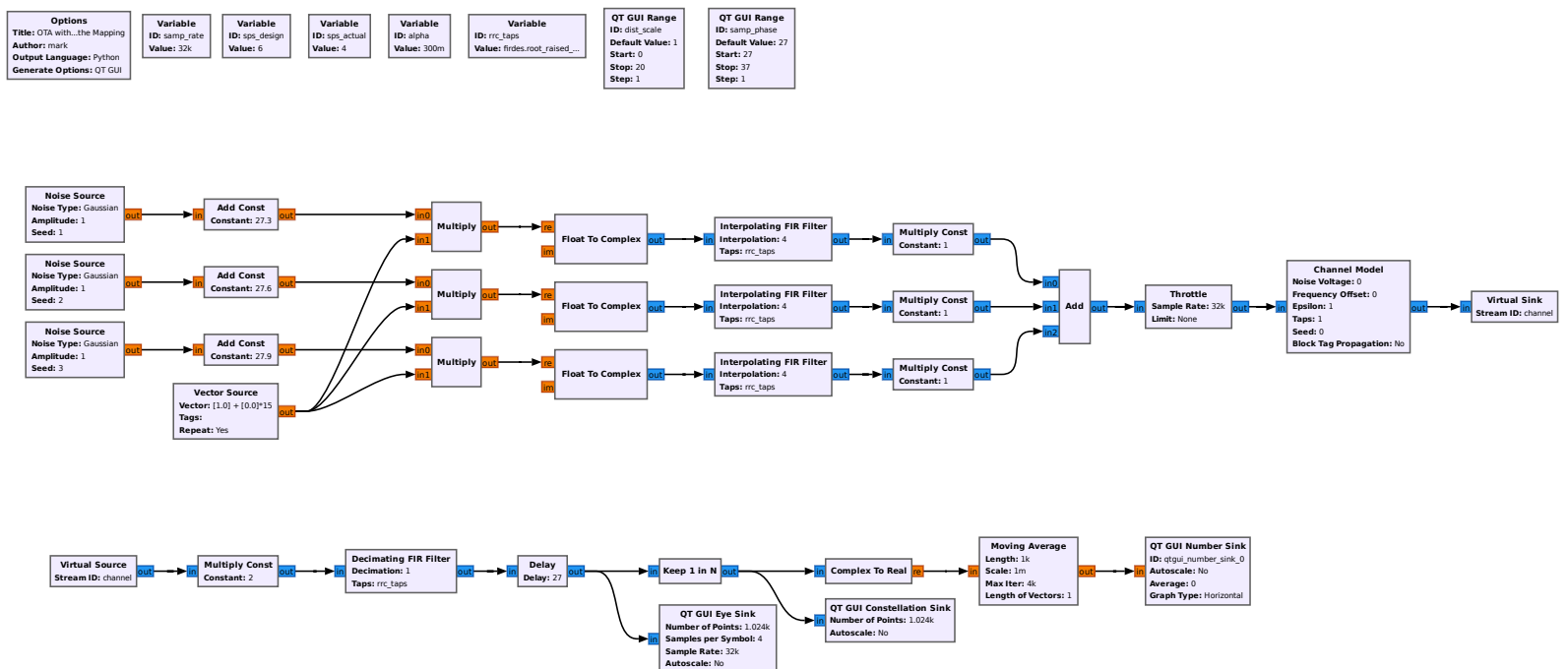


Figure 14: GNU Radio flowgraph implementing the Over-the-Air computation network over a multipath channel with controlled FIR delay taps and Root Raised Cosine pulse shaping.

multi-tap delay profile matching Eq. (5). Pulse shaping is realized using Root Raised Cosine Filter blocks with interpolation and decimation factors matched to $\text{sps_actual} = 6$. Real-time inspection via the QT GUI Eye Sink and QT GUI Constellation Sink confirms the identical signal integrity phenomena observed in MATLAB and Simulink: the multipath tap induces severe eye closure, eliminating the optimal decision strobe and dispersing constellation energy across the decision boundary.

1.f Summary of Chapter Findings

This chapter demonstrated that multipath propagation and unresolvable fractional delays constitute fundamental physical bottlenecks for Over-the-Air computation:

- The closed-form analytical derivation proved that multipath dispersion imposes an irreducible MSE error floor scaling strictly with the square of the echo attenuation (g^2).
- Farrow filter analysis established that quartic ($L = 4$) Lagrange interpolation eliminates synthetic simulation droop, guaranteeing high-fidelity discrete modeling.
- Multi-rate sample-based Simulink simulations corroborated the analytical g^2 scaling law and revealed critical implementation requirements regarding filter group delay compensation and downsampler phase alignment.
- Eye and constellation diagnostics provided intuitive visual confirmation of trajectory shearing, loss of zero-derivative strobe plateaus, and DC gain elevation.
- Multi-carrier asynchronous modeling demonstrated that independent CFOs translate baseband DC components into beat-frequency sinusoids oscillating symmetrically around zero, producing envelope beating and destroying the coherent timing margin, which can be addressed via non-coherent Euclidean magnitude rectification ($|z(t)| = \sqrt{I^2 + Q^2}$) at the expense of multi-sensor envelope fading.

These conclusions motivate the development of advanced equalization strategies, adaptive waveform designs, and robust precoding frameworks to safeguard analog computation over dispersive wireless channels.

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